



Laminar burning velocities of propionic acid + air flames: Experimental, modeling and data consistency study

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ARTICLE INFO

Article history:

Received 27 January 2021

Revised 22 March 2021

Accepted 22 March 2021

Keywords:

Propionic acid

Acetic acid

Formic acid

Burning velocity

Kinetics

Data consistency

ABSTRACT

Laminar burning velocities of propionic acid + air flames have been determined using the heat flux method at atmospheric pressure and initial gas mixture temperature of 348 K over the range of equivalence ratios 0.7–1.3. The detailed kinetic model of the authors was extended by the reactions of propionic acid and its intermediates. Attention has been paid to the proper description of the formation and consumption of 1,1-propenediol, methyl ketene, and acrylic acid. New experimental results have been compared with the modelling using this kinetic mechanism and the mechanism of Zhang et al. (2021). The comparison of the new and available from the literature measurements obtained at different initial temperatures with predictions of the two models allowed for analysis of the data consistency for the burning velocities of propionic acid, as well as for acetic and formic acids. Inconsistencies of some datasets for these short-chain carboxylic acids were identified and discussed.

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1. Introduction

The combustion chemistry of carboxylic acids has recently attracted significant interest motivating theoretical, experimental and modelling studies. Indeed, the acids are harmful atmospheric pollutants registered in the exhaust of combustion devices that prompted the development of detailed kinetic models describing their formation in the combustion of hydrocarbons [1]. For some biofuels, carboxylic acids are important primary intermediates as revealed by theoretical calculations of, e.g., Sun et al. [2]. Specifically, the formation of ethylene and parent acid has the lowest barrier in the decomposition of ethyl esters (typical components of bio-diesel) that contributes to the fuel consumption in flames and must be taken into account in interpreting shock tube experiments. Some renewable fuels produced by woody biomass pyrolysis (bio-oils) contain significant fractions of formic and acetic acids [3], therefore reactions of carboxylic acids should be considered in kinetic models designed for many alternative fuels.

The development of detailed reaction mechanisms for carboxylic acids is hampered by the lack of kinetic information, such as site-specific rate constants and very limited experimental data for model validation. For example, the only target to test the kinetic model for formic acid oxidation developed by Marshall and

Glarborg [4] was rather old measurements of the burning velocity by De Wilde and Van Tiggelen [5]. Recent experimental studies on the flame propagation in mixtures containing formic acid and different combinations of oxygen and inert [6–8] were in many cases found in good agreement with the predictions of this model. However, further analysis [9] revealed that some datasets of the burning velocities obtained at different temperatures [6] are inconsistent. Particularly, it was found that the measurements in formic acid + air flames at 423 K are better reproduced by the model of Marshall and Glarborg [4], while those at 373 K are closer to the predictions of our mechanism [9]. Recently Osipova et al. [10] extended their study of formic acid flames [8] using the heat flux method for measurements of the burning velocity of HCOOH and of HCOOH+H₂ fuel mixtures and found that the models of Sarathy et al. [6], of Marshall and Glarborg [4], and combination of the latter with Aramco 3.0 mechanism all significantly overpredict their experimental data.

The first measurements of the laminar burning velocities of acetic acid + air flames at 1 atm and initial gas temperatures of 338 K, 348 K, and 358 K were reported by Christensen and Konnov [11], and detailed reaction mechanism for acetic acid (CH₃CO₂H) combustion was developed based on the available kinetic data pertinent to acetic acid and ketene, which was recognized as the key intermediate. The model over-predicted measured burning velocities by about 3 cm/s that was attributed to the uncertainties in high-temperature chemistry of ketene since the calculated burning velocities were insensitive to the reactions of acetic acid and

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mostly governed by C1 chemistry typical for all hydrocarbons and reactions of CH_2CO . To resolve this discrepancy between the measurements and modeling, the mechanism of Christensen and Konnov [11] was elaborated via theoretical studies mostly focused on the reactions of ketene with key flame radicals [12–14]. Only slight improvement was achieved that made the authors [14] to support the opinion of Cavallotti et al. [15] that the kinetic model of acetic acid is most probably incomplete and requires consideration of deficient reactions of the CH_2COOH radical.

Recently Zhang et al. [16] presented new measurements of the laminar burning velocities of acetic acid + air mixtures at 423 K and 1 atm performed in spherical flames. They compared these results with predictions of the mechanisms of Cavallotti et al. [15], of Christensen and Konnov [11], and of their own mechanism developed based on the previous studies. Zhang et al. [16] found that the latter two models are very close to each other and to the measurements at 423 K, while overpredict the burning velocities at lower temperatures [11]. This divergence may indicate an inconsistency between two experimental datasets [11,16], which should be further analysed.

Zhang et al. [16] also performed unique measurements of the laminar burning velocities of propionic acid + air flames at 423 K and 1 atm. These experimental data were compared with predictions of their own mechanism and of the model of Sun et al. [2]. The model of Zhang et al. [16] is in better agreement with the burning velocities of propionic acid ($\text{C}_2\text{H}_5\text{COOH}$), which the authors attributed to the improved kinetics of reactions of the primary radicals $\text{CH}_2\text{CH}_2\text{COOH}$ and CH_3CHCOOH theoretically investigated in their study. On the other hand, Zhang et al. [16] reported notable experimental problems for propionic acid flames, where day-to-day disparities indicated an interaction of the acid with the surface of the combustion bomb. This effect was most prominent in lean and stoichiometric flames, where both mechanisms tested [2,16] overpredicted burning velocities, therefore independent measurements could be very helpful for the model corroboration. So far, no other flame studies of the propionic acid have been performed.

Summarising the current status of the availability of experimental data for validation of detailed kinetic mechanisms for short-chain acids one may conclude that independent measurements of the burning velocities are highly necessary. Therefore, the goal of the present work was to investigate flame propagation in propionic acid + air flames and analyse data consistency for acetic and propionic acids via comparison with experiments of Zhang et al. [16] as well as for formic acid evaluating available literature data. The heat flux method was used to determine burning velocities at 348 K and atmospheric pressure. The detailed kinetic model of the authors was extended by the reactions of propionic acid and its intermediates. The consistency of the measurements performed at different conditions was assessed through comparison with the predictions of this mechanism and of the model of Zhang et al. [16] following the approach earlier implemented for flames of hydrogen [17], syngas [18] and formic acid [9].

2. Experimental details

The burning velocity measurements have been performed at atmospheric pressure using the heat flux method. A review of the method and its development, detailed descriptions of the experimental setup and data processing algorithms have been presented elsewhere [19], thus only the relevant elements are outlined below.

All heat flux burners employed so far in our laboratory are made of brass and it is well known that propionic acid can be corrosive towards such a material. Hence, the burner with the worst performances due to an old design, designated as burner “E2” by Alekseev et al. [19] has been employed. This burner was

successfully used by Christensen and Konnov [11] and by Lubrano Lavadera and Konnov [9] to determine the laminar burning velocities of acetic and formic acid flames. In those studies [9,11], no noticeable consumption of the acids or formation of unexpected products close to the burner surface was reported. Even though absorption and desorption of acids on the internal surfaces of the plenum chamber and the burner head cannot be ruled out, these processes are believed to be balanced and in the steady state at each condition of measurements.

The experimental apparatus consists of a 2 mm-thick plate attached to the burner head and perforated with 0.5 mm-diameter holes at a pitch of 0.7 mm. The burner plate has a nominal diameter of 3 cm, but the exact value of the effective perforation area is not well defined. Therefore, to determine it, a procedure similar to that employed by Christensen and Konnov [11] has been adopted, which however leads to a higher uncertainty of the measured burning velocity than commonly reported for measurements performed with the heat flux method (usually < 1 cm/s). The burner plate is surrounded by a heating jacket supplied with thermostatic water to keep its edge at a constant temperature of 368 K, while the temperature of the fresh gas mixture (T_g) is controlled by a separate heating system in the plenum chamber, supplied with water. In the present experiments, T_g has been fixed at 348 K (± 1 K). This temperature has been chosen in order to cover a wide range of equivalence ratios, which is constrained by the vaporization limit of the mixture.

The laminar burning velocity was determined by varying the flow rate of the gas mixture until a uniform radial temperature distribution was achieved over the burner plate, recorded by eight E-type thermocouples (0.5 mm diameter) inserted into the burner plate. For this state, the net heat flux from/to the burner plate is zero and the velocity of the unburned gas is equal to the adiabatic laminar burning velocity. Hermanns [20] found that the temperature difference between the burner plate and the unburned gas mixture should be higher than 30 K to avoid flame instabilities. However, in the present experiments, this ΔT has been set to 20 K. Consequently, most of the investigated flames exhibited corrugation or cellularity around adiabatic conditions. In these cases, the burning velocity was determined by extrapolation from flat sub-adiabatic flames. An example of evaluating the burning velocity from sub-adiabatic flames and the associated uncertainties can be found in [19].

A mixing panel provides a controlled flow of the vaporized fuel and air to set the required equivalence ratio. The liquid fuel feeding system consists of a stainless steel cylinder, pressurized with nitrogen to feed the propionic acid through a Coriolis liquid mass flow controller into a Controlled Evaporator Mixer (CEM), both made by Bronkhorst High-Tech. The total uncertainty in the liquid flow rate is a sum of 0.5 g/h and the stated flow repeatability of the liquid flow controller, which corresponds to 0.2% of the reading. Propionic acid was used as delivered from Sigma Aldrich in sealed bottles. Synthetic air was delivered by Linde (21% O_2 , 1% relative uncertainty) and its required flow rates were metered utilizing a Bronkhorst High-Tech digital thermal mass flow controller. The gas flow meter was calibrated before the measurements using a MesaLabs Definer 220 positive displacement calibrator. The total uncertainty of the air flow rate is a sum of 1% stated accuracy of the calibrator and the stated flow repeatability of the mass flow controller, which corresponds to 0.2% of the set operating conditions. In order to prevent fuel condensation throughout the fueling system, an electrically heated tube was used to connect the evaporator to the plenum chamber. The temperature of the heated pipe was set equal to T_g .

Detailed analysis and quantification of the experimental uncertainties were reported earlier [19] and the overall accuracy of burning velocity in the present measurements was estimated to be

better than ± 1.8 cm/s. The equivalence ratio of the mixtures was varied from 0.7 to 1.3 with a maximum uncertainty of 0.017. All experimental data with associated uncertainties are tabulated in the Supplemental Material.

3. Modeling details

Premixed flame modelling, as well as sensitivity analysis, were performed using ANSYS Chemkin 17.0 software [21]. The laminar burning velocity was calculated considering multicomponent transport and thermal diffusion options. A grid-independent solution was ensured by setting the parameters GRAD and CURV to 0.03, resulting in a number of grid points around 500–700 over the domain of 3 cm.

As is mentioned earlier, two models were tested and compared with experimental data: the model of the authors described in the following and the mechanism of Zhang et al. [16]. The latter model was very difficult to converge using the parameters specified above. An examination of this mechanism revealed that the rate constants of several reactions in the reverse direction violate the collision limit [22]. To resolve this issue, reactions in the model of Zhang et al. [16] which rate constants exceeded 10^{20} cm³/mole s were made irreversible. These reactions are listed in the Supplemental Material. This modification does not significantly affect calculated burning velocities: a direct comparison of the calculated burning velocities using the original (for a limited number of successfully converged results) and corrected model of Zhang et al. [16] showed a difference not exceeding 0.1 cm/s.

4. Model development

4.1. Thermodynamic and transport parameters

New species introduced in the present kinetic model include propionic and acrylic (CH_2CHCOOH) acids, methyl ketene (CH_3CHCO), and 1,1-propenediol ($\text{CH}_3\text{CHC}(\text{OH})_2$), as well as their primary radicals. Thermodynamic data for $\text{C}_2\text{H}_5\text{COOH}$, CH_2CHCOOH , CH_3CHCO , and CH_2CHCO and $\text{C}_2\text{H}_5\text{C}(\text{O})\text{O}$ radicals were taken from the Burcat's database [23]. Thermodynamic data for $\text{CH}_3\text{CHC}(\text{OH})_2$, $\text{CH}_2\text{CH}_2\text{COOH}$, and CH_3CHCOOH were adopted from the study of Sun et al. [2]. Finally, thermodynamic data for CH_3CCO radical were obtained from those of CH_2CHCO radical adjusting enthalpy of formation by 10 kcal following theoretical calculations of Chin et al. [24].

Transport parameters for CH_3CHCO were adopted from the theoretical database of Jasper and Miller [25], while for methyl ketene radicals were assumed to be the same. Transport data for propionic acids were assumed equal to those of $\text{NC}_3\text{H}_7\text{O}_2$, which were taken from the LLNL kinetic model [26]. For other radicals formed from $\text{C}_2\text{H}_5\text{COOH}$ and acrylic acid, the transport parameters equal to propionic acid were assumed.

4.2. Decomposition of propionic acid

The first study of propionic acid decomposition was performed by Blake and Hole in a static reactor at sub-atmospheric pressures (0.03–0.26 atm) and within the temperature range 769–853 K [27]. They suggested that dehydration and decarboxylation



are the primary initiation reactions in the acid pyrolysis and derived their rate constants from the rates of formation of the primary products. Doolan et al. [28] investigated the decomposition

of propionic acid in a single pulse shock tube over the pressure range 14–16 atm covering temperatures from 1100 to 1500 K. They developed the first detailed kinetic model for $\text{C}_2\text{H}_5\text{COOH}$ pyrolysis and demonstrated good agreement between modelling and their measurements of the major and minor products. This model included reactions



together with reactions (1) and (2) as initiation processes. Later Chen et al. [29] performed theoretical calculations of the rate constants of dehydration and decarboxylation of propionic acid in the temperature range 700–1000 K at 0.13 atm and found them in agreement with the results of Blake and Hole [27]. Clark et al. [30] in their theoretical study of decomposition pathways of carboxylic acids included isomerisation



forming 1,1-propenediol and calculated high-pressure limit rate constants of reactions (1), (2) and (5). Finally, pressure-dependent rate constants of decomposition and isomerisation of $\text{C}_2\text{H}_5\text{COOH}$ have been calculated by Sun et al. [2] and all accepted in the present work. These rate constants are within an order of magnitude within the rate constants suggested by Doolan et al. [28] as illustrated in Fig. S1 in the Supplemental Material, although the relative importance is not the same. Specifically, Doolan et al. [28] imply that dehydration and decarboxylation prevail at temperatures below 1000 K, while theoretical calculations [2] favour reaction of decomposition (3) at all conditions, especially at high temperatures pertinent to the experimental shock tube study [28]. Available rate constants of dehydration and decarboxylation of propionic acid are summarized in Figs. S2 and S3, respectively. One may note that earlier calculations of Chen et al. [29] and the rate constant of Blake and Hole [27] dramatically deviate from the recent theoretical studies probably due to ignorance of decomposition reactions (3) and (4).

Battin-Leclerc et al. [1] suggested bimolecular recombination $\text{C}_2\text{H}_5\text{CO} + \text{OH} = \text{C}_2\text{H}_5\text{COOH}$ to explain formation of propionic acid in flames. However, the adduct is highly excited and should be thermalized; therefore, only termolecular recombination is possible. In the present work the rate constant of reaction



was estimated to be the same as suggested for acetic acid formation [11]. The rate constants suggested in the present work are summarized in Table 1, while the rate constants directly adopted from the literature (e.g. for reactions (1)–(5)) are present in Table S2 in the Supplemental Material. When pressure-dependent rate constant is provided in PLOG format, only the term at 1 atm is listed in these tables.

Decomposition of propionic acid forming H atoms is kinetically not important due to much higher barrier compared to reactions (1)–(5), however, recombination may contribute to chain termination. Therefore, for reactions



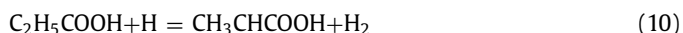
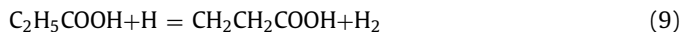
the rate constant calculated for acetic acid decomposition $\text{CH}_3\text{CO}_2\text{H} = \text{H} + \text{CH}_2\text{CO}_2\text{H}$ by Cavallotti et al. [15] was tentatively adopted in the present study.

Table 1Rate constants suggested in the present work, units are $\text{cm}^3 \cdot \text{mole}^{-1} \cdot \text{s}^{-1} \cdot \text{cal}^{-1} \cdot \text{K}$, $k = A \exp(-E_a/RT)$. UF - uncertainty factor.

No	Reaction	A	n	E_a	Temperatures, K	UF
6	$\text{C}_2\text{H}_5\text{CO} + \text{OH} + \text{M} = \text{C}_2\text{H}_5\text{COOH} + \text{M}$	$1.000\text{E}+22$	-2.0	0.0	300–2000	10
7	$\text{C}_2\text{H}_5\text{COOH} = \text{CH}_2\text{CH}_2\text{COOH} + \text{H}$ PLOG	$6.310\text{E}+58$	-13.25	120010.6	700–2100	5
8	$\text{C}_2\text{H}_5\text{COOH} = \text{CH}_3\text{CHCOOH} + \text{H}$ PLOG	$6.310\text{E}+58$	-13.25	120010.6	700–2100	5
9	$\text{C}_2\text{H}_5\text{COOH} + \text{H} = \text{CH}_2\text{CH}_2\text{COOH} + \text{H}_2$ +	$6.445\text{E}+03$ $7.345\text{E}+09$	2.93 1.31	5246.0 10315.0	300–2000	4
10	$\text{C}_2\text{H}_5\text{COOH} + \text{H} = \text{CH}_3\text{CHCOOH} + \text{H}_2$ +	$3.060\text{E}+04$ $5.734\text{E}+13$	2.8 0.1	3473.0 10569.0	300–2000	4
12	$\text{C}_2\text{H}_5\text{COOH} + \text{OH} = \text{CH}_2\text{CH}_2\text{COOH} + \text{H}_2\text{O}$	$5.250\text{E}+09$	1.08	2187.0	550–2000	3
13	$\text{C}_2\text{H}_5\text{COOH} + \text{OH} = \text{CH}_3\text{CHCOOH} + \text{H}_2\text{O}$	$4.670\text{E}+06$	1.62	-86.8	550–2000	3
14	$\text{C}_2\text{H}_5\text{COOH} + \text{OH} = \text{C}_2\text{H}_5\text{C}(\text{O})\text{O} + \text{H}_2\text{O}$	$1.230\text{E}-02$	4.337	-3235.0	500–2500	3
21	$\text{C}_2\text{H}_5\text{COOH} + \text{O} = \text{CH}_2\text{CH}_2\text{COOH} + \text{OH}$	$1.780\text{E}+00$	4.01	3420.0	250–2000	10
22	$\text{C}_2\text{H}_5\text{COOH} + \text{O} = \text{CH}_3\text{CHCOOH} + \text{OH}$	$4.550\text{E}+01$	3.60	1970.0	250–2000	10
40	$\text{CH}_2\text{CH}_2\text{COOH} + \text{H} = \text{C}_2\text{H}_5 + \text{HOCO}$	$3.700\text{E}+24$	-2.92	12505.0	300–1500	5
41	$\text{CH}_2\text{CH}_2\text{COOH} + \text{H} = \text{CH}_2\text{CHCOOH} + \text{H}_2$	$1.800\text{E}+12$	0.0	0.0	300–1500	5
42	$\text{CH}_2\text{CH}_2\text{COOH} + \text{O} = \text{CH}_2\text{CO}_2\text{H} + \text{CH}_2\text{O}$	$9.600\text{E}+13$	0.0	0.0	300–1500	5
43	$\text{CH}_2\text{CH}_2\text{COOH} + \text{OH} = \text{CH}_2\text{CHCOOH} + \text{H}_2\text{O}$	$2.400\text{E}+13$	0.0	0.0	300–1500	5
44	$\text{CH}_3\text{CHCOOH} + \text{H} = \text{H}_2\text{O} + \text{CH}_3\text{CHCO}$ PLOG	$1.051\text{E}+24$	-3.1128	7038.34	700–2100	5
45	$\text{CH}_3\text{CHCOOH} + \text{H} = \text{CO}_2 + \text{C}_2\text{H}_6$ PLOG	$1.803\text{E}+25$	-3.5494	7306.49	700–2100	5
46	$\text{CH}_3\text{CHCOOH} + \text{H} = \text{HOCO} + \text{C}_2\text{H}_5$ PLOG	$1.365\text{E}+21$	-1.9328	8941.49	700–2100	5
48	$\text{CH}_3\text{CHC}(\text{OH})_2 + \text{H} = \text{C}_2\text{H}_5 + \text{HCOOH}$	$3.000\text{E}+12$	0.0	0.0	300–2000	10
49	$\text{CH}_3\text{CHC}(\text{OH})_2 + \text{O} = \text{S}_2\text{C}_2\text{H}_5\text{O} + \text{HOCO}$	$1.000\text{E}+13$	0.0	0.0	300–2000	10
50	$\text{CH}_3\text{CHC}(\text{OH})_2 + \text{OH} = \text{C}_2\text{H}_5\text{OH} + \text{HOCO}$	$1.000\text{E}+12$	0.0	0.0	300–2000	10
77	$\text{CH}_3 + \text{HCCO} = \text{CH}_3\text{CHCO}$	$1.800\text{E}+13$	0.0	0.0	500–2000	5
79	$\text{CH}_3\text{CCO} + \text{H} = \text{CH}_3\text{CHCO}$	$1.000\text{E}+13$	0.0	0.0	500–2000	5
80	$\text{CH}_2\text{CHCO} + \text{H} = \text{CH}_3\text{CHCO}$	$1.000\text{E}+13$	0.0	0.0	500–2000	5
81	$\text{CH}_2\text{CHCO} = \text{C}_2\text{H}_3\text{CO}$	$1.000\text{E}+13$	0.0	3000.0	500–2000	5
82	$\text{CH}_3\text{CHCO} + \text{H} = \text{C}_2\text{H}_5 + \text{CO}$	$7.400\text{E}+12$	0.0	1745.0	240–440	3
83	$\text{CH}_3\text{CHCO} + \text{H} = \text{CH}_3\text{CCO} + \text{H}_2$	$8.036\text{E}+05$	2.5	12280.0	500–2000	5
84	$\text{CH}_3\text{CHCO} + \text{H} = \text{CH}_2\text{CHCO} + \text{H}_2$	$6.457\text{E}+12$	0.0	4445.0	500–2000	5
85	$\text{CH}_3\text{CHCO} + \text{O} = \text{CH}_3\text{HCO} + \text{CO}$	$2.600\text{E}+12$	0.0	-495.0	230–449	3
86	$\text{CH}_3\text{CHCO} + \text{O} = \text{C}_2\text{H}_4 + \text{CO}_2$	$2.600\text{E}+11$	0.0	-495.0	230–449	3
87	$\text{CH}_3\text{CHCO} + \text{O} = \text{CH}_3\text{CCO} + \text{OH}$	$6.000\text{E}+10$	0.7	7630.0	500–2000	5
88	$\text{CH}_3\text{CHCO} + \text{O} = \text{CH}_2\text{CHCO} + \text{OH}$	$1.800\text{E}+11$	0.7	5880.0	500–2000	5
90	$\text{CH}_3\text{CHCO} + \text{OH} = \text{CH}_3\text{CCO} + \text{H}_2\text{O}$	$1.110\text{E}+06$	2.0	1450.0	500–2000	5
91	$\text{CH}_3\text{CHCO} + \text{OH} = \text{CH}_2\text{CHCO} + \text{H}_2\text{O}$	$3.120\text{E}+06$	2.0	-300.0	500–2000	5
95	$\text{CH}_3\text{CCO} + \text{HO}_2 = \text{CH}_3\text{CHCO} + \text{O}_2$	$6.000\text{E}+11$	0.0	0.0	500–2000	5
96	$\text{CH}_2\text{CHCO} + \text{HO}_2 = \text{CH}_3\text{CHCO} + \text{O}_2$	$6.000\text{E}+11$	0.0	0.0	500–2000	5
97	$\text{CH}_3\text{CHCO} + \text{O}_2 = \text{CH}_3\text{HCO} + \text{CO}_2$	$2.000\text{E}+13$	0.0	61500.0	500–2000	5
98	$\text{CH}_3\text{CHCO} + \text{HO}_2 = \text{CH}_3\text{HCO} + \text{CO} + \text{OH}$	$6.000\text{E}+11$	0.0	12738.0	500–2000	5
99	$\text{CH}_3\text{CHCO} + \text{CH}_2 = \text{C}_3\text{H}_6 + \text{CO}$	$1.970\text{E}+08$	1.49	7925.0	300–3000	3
100	$\text{CH}_3\text{CHCO} + \text{CH}_2 = \text{C}_2\text{H}_3\text{CO} + \text{CH}_3$	$5.900\text{E}+02$	3.58	6545.0	300–3000	3
101	$\text{CH}_3\text{CHCO} + \text{CH}_3 = \text{IC}_3\text{H}_7 + \text{CO}$	$1.240\text{E}+05$	2.29	10642.0	300–3000	3
102	$\text{CH}_3\text{CHCO} + \text{CH}_3 = \text{C}_2\text{H}_3\text{CO} + \text{CH}_4$	$1.550\text{E}+02$	3.38	10512.0	300–3000	3

4.3. Abstraction from propionic acid

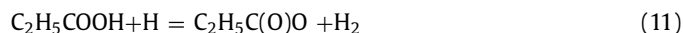
Rate constants of reactions between H atoms and propionic acid (as well as other carboxylic acids) were calculated by Mendes et al. [31]. However, Christensen and Konnov [11] noted that for reactions of OH radicals with acetic acid, where a comparison with limited experimental data was possible, these calculations underpredict the rate constant measurements. Therefore, in the present work the rate constants [31] of reactions



were compared with analogous reactions of propane. Figure 1 presents the calculations of Mendes et al. [31] for reaction (9) and $\frac{1}{2}$ of the rate constants of reaction $\text{C}_3\text{H}_8 + \text{H} = \text{NC}_3\text{H}_7 + \text{H}_2$ from the review of Tsang [32] and from the theoretical study of Sivaramakrishnan et al. [33]. Similarly, Figure 2 depicts the calculations of Mendes et al. [31] for reaction (10), and the rate constants of reaction $\text{C}_3\text{H}_8 + \text{H} = \text{IC}_3\text{H}_7 + \text{H}_2$ [32,33]. In both cases, the recommen-

dations of Tsang [32] are close to the theoretical expressions of Sivaramakrishnan et al. [33] and significantly higher than the rate constants of Mendes et al. [31]. In the present work, the rate constants based on the analogy with C_3H_8 [33] are adopted.

For the abstraction of H from acidic group



Sumathi et al. [34] suggested a generic rate constant which is compared in Fig. S4 in the Supplemental Material with the calculations by Mendes et al. [31] and with the theoretical expression for reaction $\text{CH}_3\text{CO}_2\text{H} + \text{H} = \text{H}_2 + \text{CH}_3\text{CO}_2$ reported by Cavallotti et al. [15]. This reaction has a very minor impact on propionic acid oxidation in flames since its rate constant is more than an order of magnitude smaller than those of reactions (9) and (10). In the present study, the calculations of Mendes et al. [31] are adopted, which are close to the generic rate [34] at high temperatures similar to the comparison performed for parent acetic acid reactions [11].

The analogy of the rate constants of H atom abstraction from propionic acid and propane could be validated for reactions with

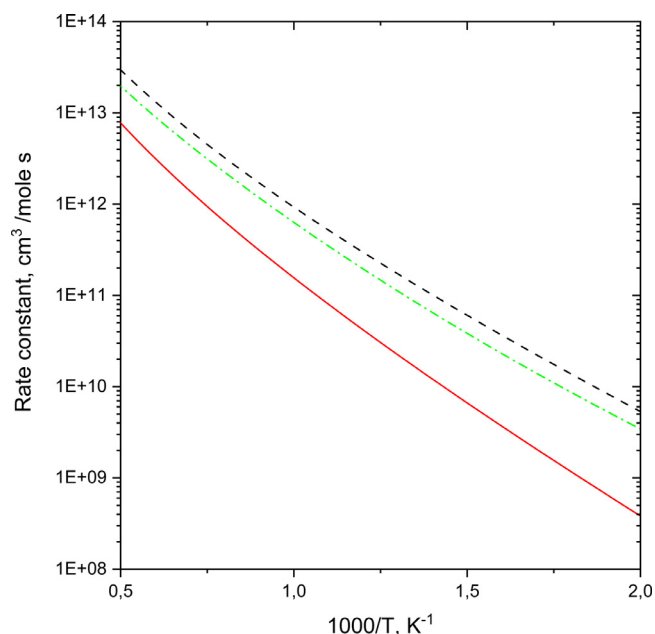


Fig. 1. Rate constants relevant to reaction $\text{C}_2\text{H}_5\text{COOH} + \text{H} = \text{CH}_2\text{CH}_2\text{COOH} + \text{H}_2$. Red solid line: calculations of Mendes et al. [31], dash black line: $\frac{1}{2}$ of the abstraction from C_3H_8 [32], green dash-dot line: $\frac{1}{2}$ of the abstraction from C_3H_8 [33]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

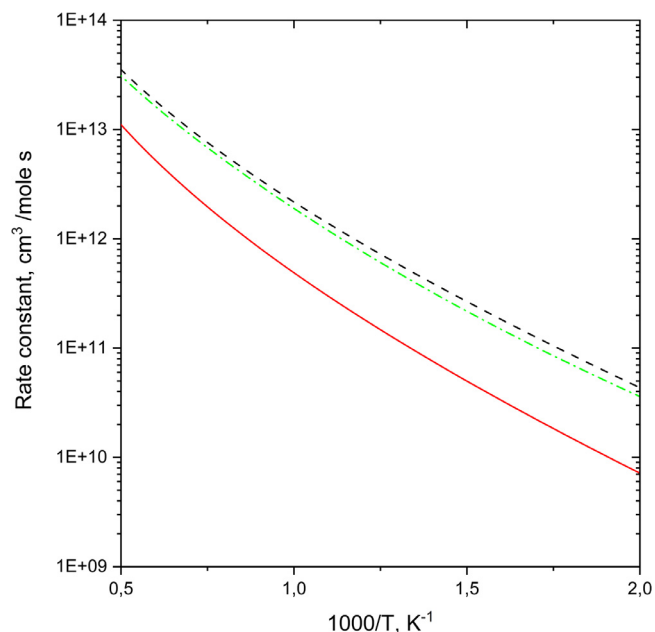


Fig. 2. Rate constants relevant to reaction $\text{C}_2\text{H}_5\text{COOH} + \text{H} = \text{CH}_3\text{CHCOOH} + \text{H}_2$. Red solid line: calculations of Mendes et al. [31], dash black line: abstraction from C_3H_8 [32], green dash-dot line: abstraction from C_3H_8 [33]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

OH for which some experimental data are available [35–37]. Figure 3 presents the measurements of the total rate constant at room [35] and elevated temperatures shown as individual data-points [37] or as Arrhenius expression [36] up to 440 K. Also depicted are the rate constant of reaction

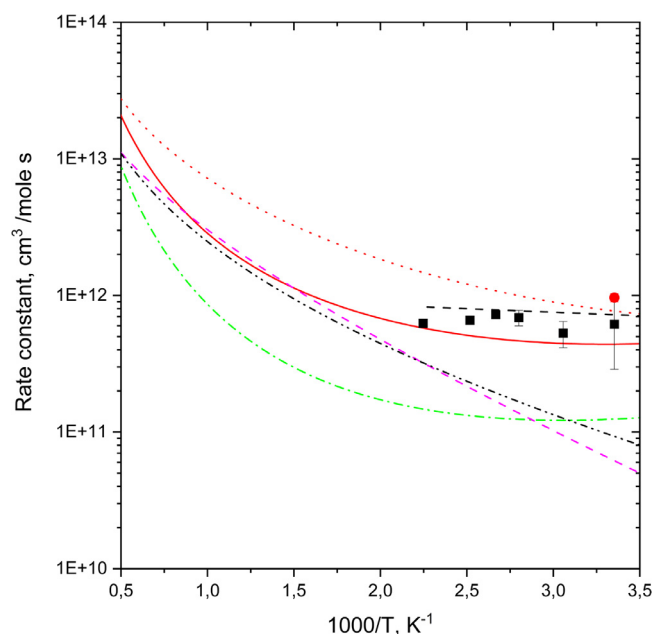
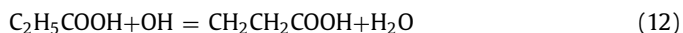
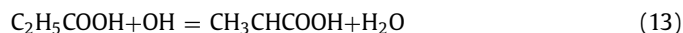


Fig. 3. Rate constants relevant to reaction $\text{C}_2\text{H}_5\text{COOH} + \text{OH} = \text{CH}_2\text{CH}_2\text{COOH} + \text{H}_2\text{O}$. Symbols and black dash line: experiments [35–37]. Solid red line: total rate constant [31], green dash-dot line: this reaction [31], dash-double dot black line: $\frac{1}{2}$ of the abstraction from C_3H_8 [38], magenta dash line: $\frac{1}{2}$ of the abstraction from C_3H_8 [39], red dot line: total rate constant adopted in this study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

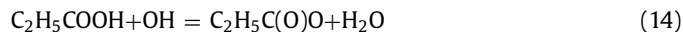
and the total rate constant of the reaction between propionic acid and OH calculated by Mendes et al. [31]. The latter is below the experimental data by about a factor of 1.5–2 similar to the calculations performed for acetic acid [11]. The rate constant predicted for reaction (12) [31] could be compared with $\frac{1}{2}$ of the expressions suggested for abstraction $\text{C}_3\text{H}_8 + \text{OH} = \text{NC}_3\text{H}_7 + \text{H}_2\text{O}$ by Cohen [38] and by Dames et al. [39]. The curvature of this rate constant in Arrhenius coordinates is much more pronounced compared to the reaction with propane, which is unexpected due to very similar bond dissociation energies in methyl groups of $\text{C}_2\text{H}_5\text{COOH}$ and C_3H_8 [40].

Figure 4 presents the rate constants relevant to reaction



in a similar way to Fig. 3. It is seen that this reaction contributes about 50% to the total abstraction by OH according to the calculations of Mendes et al. [31] from low to high temperatures. The rate expressions for reaction $\text{C}_3\text{H}_8 + \text{OH} = \text{IC}_3\text{H}_7 + \text{H}_2\text{O}$ by Cohen [38] and by Dames et al. [39] are close to each other and somewhat higher than the rate constant of reaction (13). For this and reaction (12) the rate constants based on the analogy with C_3H_8 [39] are adopted in the present work.

The rate constants relevant to reaction



are shown in Fig. 5. Together with the theoretical calculations of Mendes et al. [31], the rate constant adopted for $\text{CH}_3\text{CO}_2\text{H} + \text{OH} = \text{H}_2\text{O} + \text{CH}_3\text{CO}_2$ and discussed by Savchenkova et al. [14] is presented. As in many other cases analyzed before [11, 14] and illustrated in Figs. 1–5, predicted rate constants [31] are lower than other evaluations based on the analogy with similar functional groups. Also shown in Figs. 3–5 is the total rate constant adopted in this study, which for reaction (14) includes the same rate constant as for acetic acid. This total rate nicely agrees

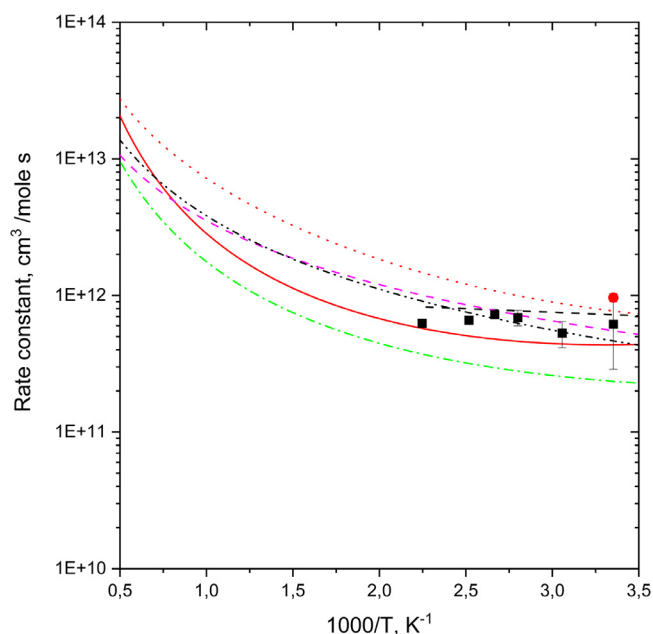


Fig. 4. Rate constants relevant to reaction $\text{C}_2\text{H}_5\text{COOH} + \text{OH} = \text{CH}_3\text{CHCOOH} + \text{H}_2\text{O}$. Symbols and black dash line: experiments [35–37]. Solid red line: total rate constant [31], green dash-dot line: this reaction [31], dash-double dot black line: abstraction from C_3H_8 [38], magenta dash line: abstraction from C_3H_8 [39], red dot line: total rate constant adopted in this study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

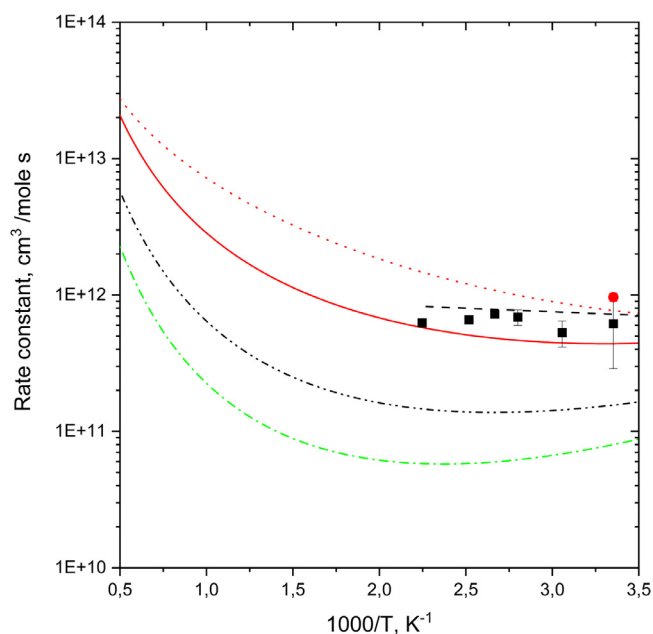
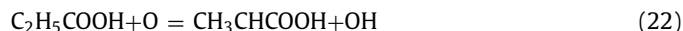
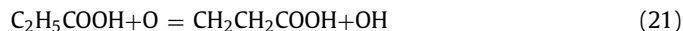


Fig. 5. Rate constants relevant to reaction $\text{C}_2\text{H}_5\text{COOH} + \text{OH} = \text{C}_2\text{H}_5\text{C}(\text{O})\text{O} + \text{H}_2\text{O}$. Symbols and black dash line: experiments [35–37]. Solid red line: total rate constant [31], green dash-dot line: this reaction [31], dash-double dot black line: abstraction from $\text{CH}_3\text{CO}_2\text{H}$ [14], red dot line: total rate constant adopted in this study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

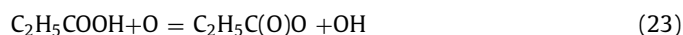
with the room temperature measurements [35–37] exceeding experimental data at 440 K by about a factor of 2 thus indicating a possible uncertainty of the approach based on the analogy with propane and with $\text{CH}_3\text{CO}_2\text{H}$. On the other hand, the stated uncertainty of the theoretical calculations of Mendes et al. [31] is a factor of 2.5, therefore the rate constants of propionic acid re-

actions with HO_2 (reactions (15)–(17)) and CH_3 radicals (reactions (18)–(20)) [31] were accepted without modifications.

For two channels of reaction between propionic acid and O atoms



Sun et al. [2] proposed to adopt the rate constants calculated by Tan et al. [41] for methyl propanoate divided by 10. They motivated this modification by the need to make them consistent with the calculations performed by Mendes et al. [31]. However, a direct comparison of the modified rate constants with those for abstraction from propane or acetic acid shows that they are lower than these analogous reactions by a factor of 10. Therefore, the rates of Tan et al. [41] are accepted here without change. The rate constant for reaction

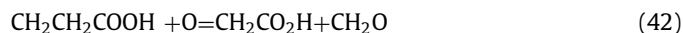


was assumed by Sun et al. [2] equal to that for reaction $\text{CH}_3\text{CO}_2\text{H} + \text{O} = \text{CH}_3\text{CO}_2 + \text{OH}$, which was in turn suggested by Christensen and Konnov [11]. For reactions with O_2 , C_2H_5 and C_2H_3 Sun et al. [2] implemented the rate constants taken from the ethyl propanoate mechanism developed by Yang et al. [42]; they all are accepted in the present work.

4.4. Reactions of primary radicals

Zhang et al. [16] contributed to the development of a propionic acid kinetic model with theoretical calculations on the $\text{C}_3\text{H}_5\text{O}_2$ energy surface. They calculated many important rate constants of decomposition and isomerisation of $\text{CH}_2\text{CH}_2\text{COOH}$, CH_3CHCOOH and $\text{C}_2\text{H}_5\text{C}(\text{O})\text{O}$ radicals all adopted in the present model.

Other reactions of $\text{CH}_2\text{CH}_2\text{COOH}$ radicals

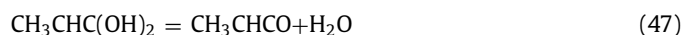


are introduced in the present work with the rate constants estimated by analogy with corresponding reactions of NC_3H_7 . Pressure dependent rate constants of $\text{CH}_2\text{CO}_2\text{H}$ radicals with H atoms were calculated by Cavallotti et al. [15] and attributed here to the similar reactions of CH_3CHCOOH

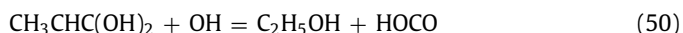
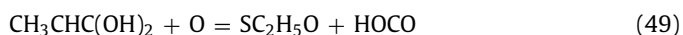


4.5. Reactions of 1,1-propenediol

1,1-propenediol, $\text{CH}_3\text{CH}(\text{OH})_2$ could be formed via isomerisation of propionic acid in reaction (5). The models of Zhang et al. [16] and Sun et al. [2] include only decomposition of this species

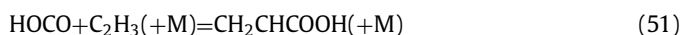


with the rate constant calculated by Sun et al. [2]. Kinetically this means that 1,1-propenediol serves as a steady-state reservoir between C_2H_5COOH and methyl ketene present in the system until these components are not completely consumed. However, 1,1-propenediol can react with major flame radicals similar to the reactions of ethenediol formed in acetic acid flames. Christensen and Konnov [11] suggested that radicals may insert barrierlessly into C=C bond with an immediate decomposition of adducts. In the present work the estimated rate constants [11] are attributed to the following reactions:

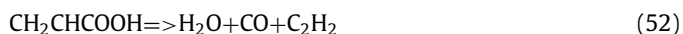


4.6. Reactions of acrylic acid

Acrylic acid could be formed from the primary radicals of propionic acid losing H atom, e.g. in reactions (41) and (43). Simplified sub-mechanism for acrylic acid oxidation was developed by Namysl et al. [40] in which primary radicals for acrylic acid are not introduced assuming immediate decomposition. Namysl et al. [40] suggested reactions of decomposition and assigned the rate constant for reaction



equal to the analogous reaction of 1-butene. For reactions of dehydration and decarboxylation



Namysl et al. [40] assigned the same rate constants as for butanoic acid. Most of the abstraction reactions (54)–(73) from acrylic acid by flame radicals are written one-way [40] and accepted here. In addition to reactions of decomposition and isomerisation on the $C_3H_5O_2$ energy surface, Zhang et al. [16] calculated the rate constants of reactions



also included in the acrylic acid sub-mechanism in the present work.

4.7. Decomposition of methyl ketene

One of the products observed during pyrolysis of propionaldehyde was methyl ketene [43] therefore simplified sub-mechanism for its oxidation was included in our earlier model [44]. The thermal decomposition [45], as well as dissociative photoionization of methyl ketene [46], have been studied experimentally, however, no rate constants were determined. Hence, that mechanism [44] was developed based on the analogy with reactions of ketene. For instance, the decomposition of methyl ketene into ethylene and carbon monoxide



was written as a pressure dependent reaction by analogy with ketene. The same approach was followed in other models [2,47],

however, later Dames et al. [48] criticized the assumption that the rate parameters for the reaction $C_2H_4 + CO \rightarrow CH_3CHCO$ could be estimated by those for the reaction of $CH_2 + CO \rightarrow CH_2CO$ and attributed the analogy to reaction



excluding reaction (76) from their model. Akih-Kumgeh and Bergthorson [49] assigned the rate constant of propylene decomposition by Chappell and Shaw [50] to the reverse reaction (–77); following this approach a temperature-independent rate constant is assumed for reaction (77). Moreover, the theoretical study of Savchenkova et al. [13] revealed that reaction (76) is pressure independent and also identified another channel of methyl ketene decomposition forming ketene and singlet methylene. This pressure-dependent reaction is included in the model in the reverse direction as



Other initiation reactions are included for completeness in the reverse direction



with estimated rate constants to allow for chain termination in the presence of H atoms.

4.8. Abstraction from methyl ketene

Abstraction from CH_3CHCO can produce 2 radicals: CH_3CCO and CH_2CHCO . 1,2 acroylyl, CH_2CHCO ($(\cdot)CH_2-CH=C=O$) has a similar structure to 1,3 acroylyl, C_2H_3CO ($CH_2=CH-C(\cdot)=O$). The theoretical analysis of Winter et al. [51] showed that they are separated by a barrier of 9.5 kJ/mol, therefore reaction



with estimated rate constant is included in the present model.

To assess the rate constants of radicals with methyl ketene an analogy with C_3H_6 was considered. Then abstraction forming CH_3CCO is assumed similar to the formation of TC_3H_5 , while the formation of CH_2CHCO is similar to C_3H_5 . Also, radicals may insert into C=C bond to form other products; for them, the analogy with ketene was used. The insertion reactions are considered to be faster than abstraction, therefore the rate constant between methyl ketene and H atoms measured by Umemoto et al. [52] is attributed to reaction



while the rate constants of reactions



are taken equal to those of propylene.

Washida et al. [53] measured the rate constant of methyl ketene with O atoms and considered two major channels forming acetaldehyde + CO and ethylene + CO_2 .



Experimentally, however, they did not detect ethylene in the products, while CH_3HCO and CH_2CO were observed. Washida et al. [53] discussed the possibility of CH_2CO formation from hot acetaldehyde or acetyl radicals. In the present work, 90% of the rate constant [53] was attributed to the channel (85) forming CO. The rates of abstraction reactions

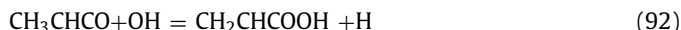


are estimated based on the analogy with C_3H_6 .

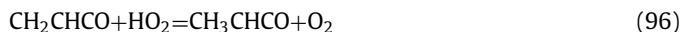
Hatakeyama et al. [54] experimentally determined the rate constant with OH which is attributed here to reaction



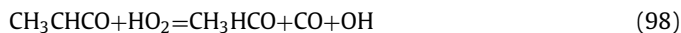
Abstraction reactions (90) and (91) should also be relatively fast and are estimated based on the analogy with C_3H_6 . In addition, Zhang et al. [16] calculated pressure-dependent rate constants of reactions



all accepted in the present model. The rate constants of initiation reactions with O_2



were estimated in the reverse direction, as well as for reactions



assuming to be equal to those of corresponding reactions with ketene evaluated by Christensen and Konnov [11].

The rate constants of ketene with CH_2 and CH_3 radicals have been theoretically calculated by Savchenkova et al. [13] and by Semnikhin et al. [12], respectively. In the present work, they are attributed to reactions



5. Burning velocities of carboxylic acids

5.1. Propionic acid

The laminar burning velocities of propionic acid + air flames measured in the present work at 348 K and atmospheric pressure are shown in Fig. 6. The maximum values are found at an equivalence ratio between $\phi = 1$ and 1.1. The measurements of Zhang et al. [16] for these flames at 423 K and 1 atm presented in Fig. 6 suggest the maximum at $\phi = 1.1$. Also shown in Fig. 6 are predictions of the model of Zhang et al. [16] and of the present

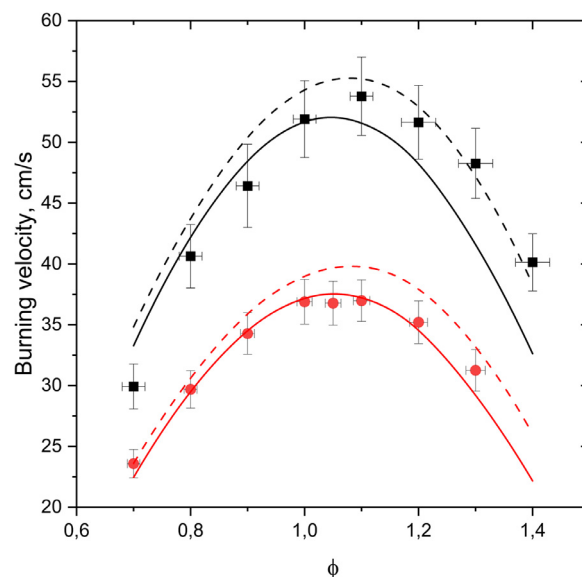


Fig. 6. Laminar burning velocities of $\text{C}_2\text{H}_5\text{COOH}$ + air flames at 348 and 423 K at 1 atm. Symbols: experiments, lines: modelling. Black squares and dashed lines: [16]; red circles and solid lines: present work. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mechanism. The latter nicely reproduces our new experimental data, while the mechanism of Zhang et al. [16] overpredicts them especially for moderately rich flames. On the other hand, both models tend to predict higher and lower burning velocities than the measurements at 423 K [16] in lean and rich mixtures, respectively.

This comparison of the experimental results obtained at two different temperatures using two different models provides some hints on the consistency of the data obtained in the two studies. A major premise of this analysis is that each model keeps its performance with respect to the measurements at different temperatures for the same fuel + air mixtures. The temperature dependence of the burning velocities can be described by simple empirical correlation $S_L/S_{L0} = (T/T_0)^\alpha$ which relates the burning velocity, S_L , at the initial temperature T , with the burning velocity, S_{L0} , at a reference temperature T_0 . Power exponents, α , then could be easily derived as a slope of the $\log(S_L)$ vs. $\log(T)$ trends from experimental data or model predictions. Usually, the power exponents vary moderately with the equivalence ratio, ϕ , attaining a minimum in the mixtures possessing maximum burning velocity [55]. An important characteristic of the temperature dependence analysis is that the power exponents predicted by different models are very close to each other even when the calculated burning velocities diverge, as was first found by Christensen et al. [56] for methyl formate flames and later demonstrated for many fuels [55].

This insensitivity of the power exponents to the mechanism used has been validated in the present work by plotting the experimental data and predictions of the two models tested in log-log scale as illustrated in Fig. S5 in the Supplemental Material. Indeed, for each equivalence ratio, variations of the calculated burning velocities in these coordinates are parallel for both mechanisms, which means that experimental data, e.g. at a lower temperature can be scaled up to the higher temperature by the factor calculated using either of the models tested. Since the present mechanism is in good agreement with the results at 348 K at least from lean to moderately rich flames, the expected values of the burning velocity at 423 K should be close to the calculated ones shown as a corresponding solid line in Fig. 6. The measurements of Zhang et al. [16] actually agree with this model predictions in

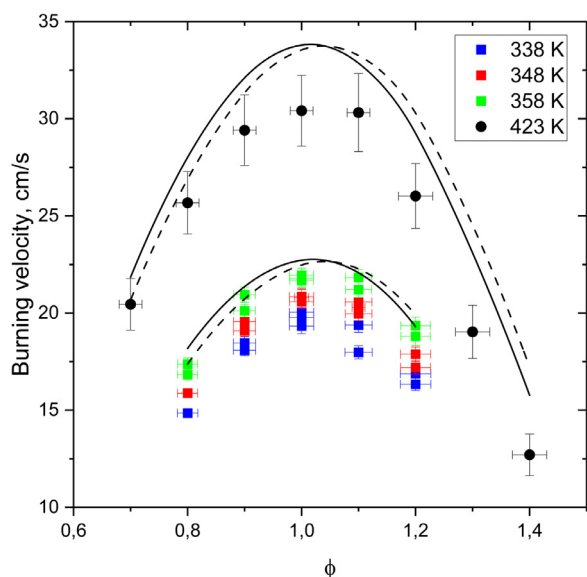


Fig. 7. Laminar burning velocities of $\text{CH}_3\text{CO}_2\text{H}$ + air flames at 338–358 and 423 K at 1 atm. Symbols: experiments, lines: modelling at 338 and 423 K. Black circles and dashed lines: [16]; squares [11], solid lines: present work.

lean and around stoichiometric mixtures even though they are visually shifted with respect to the calculations along the x-axis. The apparent shift is most probably explained by the absorption of the propionic acid on the surface of the combustion bomb as discussed by the authors [16]. They reported notable day-to-day variations of the measured burning velocities especially in lean flames, however, the absorption undoubtedly occurs for any mixture composition leading to effective leaning at all conditions. One may conclude that the present measurements of the burning velocity of propionic acid + air flames do not contradict the results of Zhang et al. [16] and could be consistent with them at least for the maximum values and the shape of the dependence vs. equivalence ratio.

Zhang et al. [16] noticed that the maximum of the burning velocity of $\text{C}_2\text{H}_5\text{COOH}$ + air flames in their experiments is found in slightly richer mixtures compared to the flames of acetic acid discussed in the following, and performed sensitivity analysis to explain this difference. The present mechanism, however, predicts the maximum in leaner flames. To reveal the disparities between key reaction controlling flame propagation in the two models, the sensitivity analysis has been performed for lean, stoichiometric, and rich flames. The sensitivity spectra shown in Fig. S6 in the Supplemental Material are actually very similar to those presented by Zhang et al. [16]. At the conditions visited, no reactions pertinent to the fuel or its primary intermediates (1,1-propenediol, methyl ketene, and acrylic acid) were found among the first 30 sensitive reactions. Important role of C_2H_x chemistry is confirmed in agreement with the earlier analysis [16]. The divergence in the behaviour of the two models seen in Fig. 6 should, therefore, be attributed to the fundamental C1-C2 chemistry, which is also manifested in the flame modelling of acetic and formic acids discussed below.

5.2. Acetic acid

In the same study, Zhang et al. [16] determined the burning velocity of acetic acid + air flames at 423 K and atmospheric pressure. Their results are presented in Fig. 7 together with the earlier measurements at initial gas temperatures of 338 K, 348 K, and 358 K using the heat flux method [11]. For the latter data, original experimental uncertainties are shown without additional error bars of ± 2 cm/s discussed by Christensen and Konnov [11] to

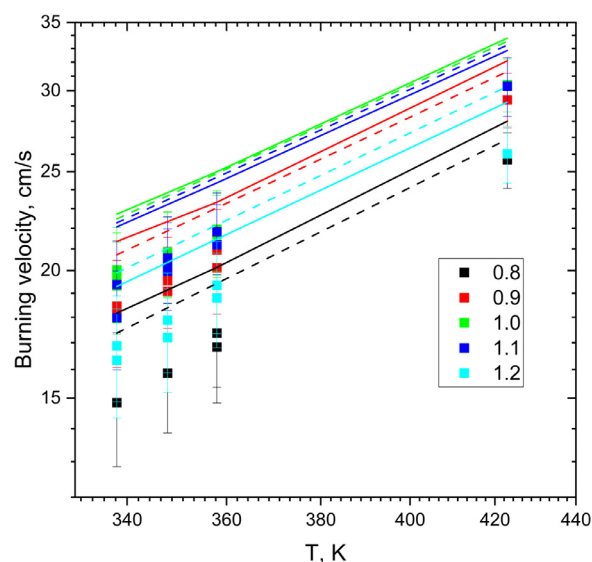


Fig. 8. Laminar burning velocities of $\text{CH}_3\text{CO}_2\text{H}$ + air flames at 338–358 [11] and 423 K [16] at 1 atm. Symbols: experiments, lines: modelling. Dashed lines: model of Zhang et al. [16]; solid lines: present mechanism.

avoid cluttered graphs. Zhang et al. [16] compared these results with predictions of the mechanisms of Cavallotti et al. [15], of Christensen and Konnov [11], and of their own mechanism and concluded that the latter two models “could both reasonably predict” the burning velocities from 338 to 423 K. However, calculated burning velocities using both the model of Zhang et al. [16] and the present mechanism depicted in Fig. 7 are systematically higher than the low-temperature group of data by 2.5–3 cm/s and agree with the results at 423 K only for lean mixtures. It is therefore instructive to analyse the data consistency of the available measurements using the same approach as used for propionic acid flames.

The burning velocities of acetic acid + air flames both determined experimentally [11,16] and predicted by the two models in log-log scale are presented in Fig. 8. Again, the calculated trends of the burning velocity increase with temperature are parallel for both mechanisms, confirming their close performance at all conditions seen in Fig. 7. However, due to relatively high experimental uncertainties (the error bars of ± 2 cm/s [11] are shown in Fig. 8) the trendlines of the measured values cannot be conclusively confronted with the modelling. Even though the slopes predicted by the two models seem somewhat lower than those indicated by the measurements, the two datasets [11,16] do not contradict each other. Nevertheless, it should be admitted that the results at temperatures 338–358 K might be lower than real values and thus independent experiments are highly necessary.

5.3. Formic acid

The analysis of the data consistency presented above for propionic acid and acetic acid flames has already been implemented for formic acid flames [9]. The burning velocities of the same mixtures HCOOH + air have been measured at 373 and 423 K in spherical flames at PRISME, Orleans [6], that allowed for direct examination of their temperature dependence. It was found [9] that these datasets are inconsistent because the increase of the burning velocity derived from the experimental data largely exceeded that predicted by two models. However, this analysis did not reveal which of these measurements are suitable for model validation since the results for formic acid + air flames at 423 K were better reproduced by the model of Marshall and Glarborg [4], while those at 373 K were closer to the predictions of our mechanism [9].

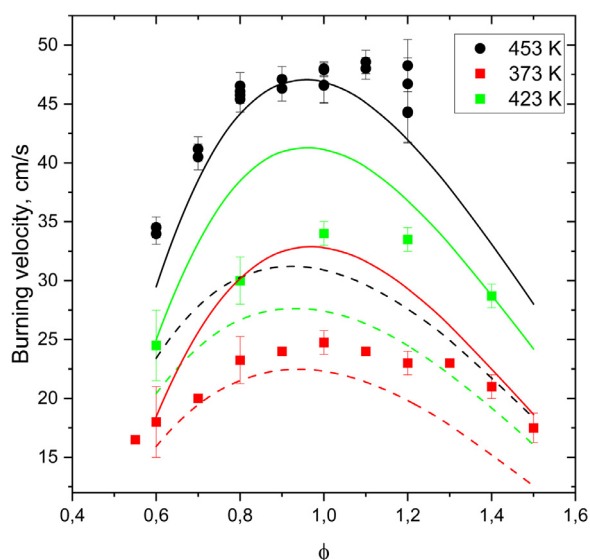


Fig. 9. Laminar burning velocities of HCOOH + O₂ + N₂ flames at 373, 423 and 453 K at 1 atm. Symbols: experiments, lines: modelling. Black circles [7], squares [10], dashed lines: model of Zhang et al. [16], solid lines: present mechanism.

Most recent measurements of Osipova et al. [10] using the heat flux method have been performed in HCOOH + O₂ + N₂ mixtures with 30% of oxygen in the oxidizer which were previously investigated in spherical flames by Yin et al. [7]. This enables further analysis and comparison of the experimental data obtained at 373 and 423 K [10] and 453 K [7] as shown in Fig. 9. Also presented in this figure are predictions of the present mechanism and the model of Zhang et al. [16]. Although the latter model was not tested by the authors for formic acid flames, they reported that the sub-mechanism of HCOOH was taken from Marshall and Glarborg [4]. The behaviour of the two mechanisms tested diverges dramatically; the model of Zhang et al. [16] underpredicts all the measurements shown in Fig. 9 only approaching the results at 373 K in lean mixtures. Instead, the present mechanism satisfactorily reproduces the burning velocities at 453 K [7] and largely overpredicts the datasets of Osipova et al. [10].

When the same experimental data and calculations using the two models are plotted in a log-log scale, it becomes obvious that the measurements performed at different temperatures are inconsistent as shown in Fig. 10. Both mechanisms predict very close variations of the burning velocity with the temperature at each equivalence ratio, while experimental trends never follow them. It should be noted that the results obtained at 453 K in spherical flames were found in good agreement with predictions of the models of Marshall and Glarborg [4], and of Lubrano Lavadera and Konnov [9] for different compositions of the oxidizer (30–40 % O₂) from moderately lean to moderately rich mixtures deviating only in very rich flames where large experimental scattering was observed, see Fig. 9. It could therefore be concluded that the measurements reported by Osipova et al. [10] contradict other datasets reported in the literature.

The present kinetic mechanism demonstrates good performance in predicting burning velocities of C₂H₅COOH, CH₃CO₂H, and HCOOH flames. It can certainly be further improved via better quantification of the rate constants pertinent to propionic acid listed in Table 1, as well as for acetic and formic acids discussed earlier [9,14]. However, the analysis of the data consistency implemented herein does not rely on the accuracy of one or another model; instead, calculations only indicate expected variations of the burning velocity with temperature allowing for comparison of

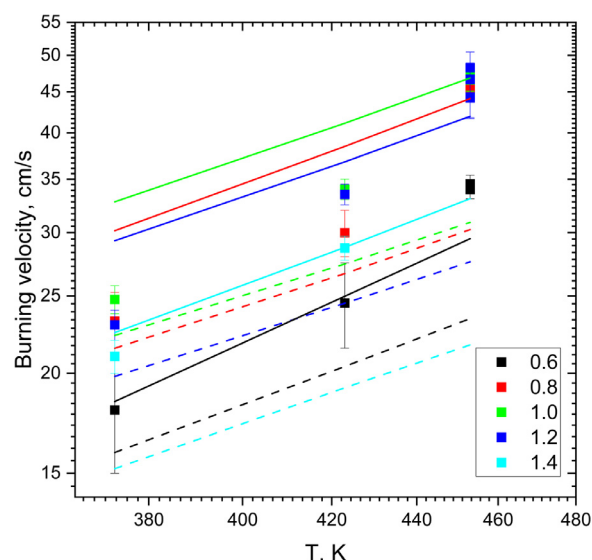


Fig. 10. Laminar burning velocities of HCOOH + O₂ + N₂ flames at 373, 423 and 453 K at 1 atm. Symbols: experiments, lines: modelling. Dashed lines: model of Zhang et al. [16]; solid lines: present mechanism.

the measurements performed at different conditions and point out on the possible outliers to be revisited experimentally.

There could be several experimental problems responsible for the disparities identified in the present analysis. The most important is probably the interaction of the acid investigated with the elements of the equipment. For instance, Zhang et al. [16] observed remarkable effect of the propionic acid adsorption on the surface of the combustion bomb, while for acetic acid the day-to-day variations in their results were within the experimental uncertainties. In the present work, the same burner was used as in the studies of Christensen and Konnov [11] and Lubrano Lavadera and Konnov [9] to determine the laminar burning velocities of acetic and formic acid flames, respectively. Although all short-chain acids may react with the burner material, only for acetic acid a notable formation of patina was observed. Sometimes, experimental difficulties occur for the fuels, which are supposed to be neutral with respect to the elements of the equipment, as was observed in the investigation of the burning velocities of acetone [57]. Therefore, revisiting the same conditions using different experimental methods is highly important to establish reliable datasets and resolve the inconsistencies identified in the present work.

6. Conclusions

The new measurements of the laminar burning velocities of propionic acid + air flames have been performed using the heat flux method at atmospheric pressure and initial mixture temperature of 348 K over the range of equivalence ratios 0.7–1.3. The detailed kinetic model of the authors previously developed for acetic and formic acid combustion was extended by the reactions of propionic acid and its intermediates. It was found in good agreement with the present results. The comparison of the new and available from the literature measurements obtained at different initial temperatures with predictions of the mechanism of Zhang et al. [16] and of the present model allowed for analysis of the data consistency for the burning velocities of propionic acid, as well as for acetic and formic acids.

1. Propionic acid. The present measurements of the burning velocity of propionic acid + air flames do not contradict the results of Zhang et al. [16] and could be consistent with them at least for the maximum values and the shape of the dependence vs.

equivalence ratio; the latter could be affected by large uncertainty in equivalence ratio due to possible fuel adsorption.

- Acetic acid. The two datasets of the burning velocity of acetic acid + air flames [11,16] do not contradict each other. The results at temperatures 338–358 K [11] might be lower than real values and thus independent experiments are highly necessary.
- Formic acid. Summarising previous and the present analyses of the data consistency, one may note that the measurements of De Wilde and Van Tiggelen [5], as well as of Sarathy et al. [6] are internally inconsistent as discussed earlier [9]. The results reported by Osipova et al. [10] contradict other datasets available in the literature, e.g. the burning velocities obtained in spherical flames by Yin et al. [7]. The measurements of Osipova et al. [8] performed in flames diluted by argon are yet to be analysed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Knut and Alice Wallenberg Foundation through grant KAW2019.0084 COCALD.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.111431.

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